Bansilal Ramnath Agarwal Charitable Trust's

VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE

[An autonomous Institute affiliated to Savitribai Phule University] Bibwewadi and Kondhwa campuses

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Unit 2: Home Assignment

1 Compute Binomial probabilities for product defect rate scenarios.
2 Solve Z-score problems and interpret outcomes for admission/eligibility criteria.
3 Perform Normal Approximation to Binomial & compare results

1. Example taken in classroom-

An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

Mean (μ) = 800 hours
Standard deviation (σ) = 40 hours
P(778 < X < 834)

For X = 778:

$$Z_1 = \frac{778 - 800}{40} = \frac{-22}{40} = -0.55$$

For X = 834:

$$Z_2 = \frac{834 - 800}{40} = \frac{34}{40} = 0.85$$

From standard Z-table:

- P(Z < 0.85) = 0.8023
- P(Z < -0.55) = 0.2912

$$0.8023 - 0.2912 = 0.5111$$

2. An entrance exam has: Mean score $\mu = 70$
Standard deviation $\sigma = 10$

A college sets eligibility cutoff at **85 marks**. Find probability that a student qualifies.

$$Z_1 = \frac{85 - 70}{10} = 1.5$$

P(Z<1.5)=0.9332

P(X>85)=1−0.9332=0.0668

<u>Conclusion-</u>

Only 6.68% students qualify
Cutoff is high and selective Useful
for:
Competitive exams
Scholarship eligibility

Z-score helps compare individual scores with population. It standardizes data for decision-making

3. $p = 0.1$ Find: $P(X \leq 15)$

$np = 10 \geq 5$ $n(1-p) = 90 \geq 5$
$\sigma = \sqrt{np(1-p)} = 3$

$P(X \leq 15) \rightarrow P(X \leq 15.5)$

$P(Z < 1.83) = 0.9664$

Method	Probability
Binomial (Exact)	0.9513
Normal Approx	0.9664

<u>Conclusion-</u>

Normal approximation simplifies binomial calculations
Accuracy improves with larger sample size
Continuity correction is essential
Suitable for real-world scenarios like: Defect rates, Survey results